

Binary Tree Representation & Types Data Structures

> **Definition:** A **Binary Tree** is a hierarchical tree data structure in which every node has at most **two children**, referred to as the **left child** and the **right child**.

1. Detailed Technical Explanation

1. Types of Binary Trees

1. FULL

3. PERFECT BINARY TREE

4. DEGENERATE / SKEWED TREE

[1]

1. **Full (Proper/Strict) Binary Tree:** Every node has either 0 or 2 children. No node has only 1 child.

2. **C**
gaps.

2. Binary Tree Representations in Memory

1. Sequential (Array-Based) Representation

For a complete binary tree stored in array `A` (1-indexed):

- No

2. Linked (Pointer-Based) Representation in C

```
```c
```

#include

```
// Structure for a Binary Tree Node
```

struct

```
// Function to create a new tree node
```

struct

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### 3. Mathematical Properties of Binary Trees

- Maximum nodes at level  $i$  (root at level 0) =  $2^i$ .

- Ma

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### 4. Must-Write Points for Exams

- In a Full Binary Tree, no node has degree 1.

- In a

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### 5. Quick Recall Flow

```
```
```

Binary